Throwing Blind: How Experience and Proprioception Support Spatial Action Without Vision

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Spring 2025

1 Introduction

Vision is one of the primary ways we interact with the world. Often, it's the first sense we rely on to perceive distance, orientation, and movement–whether we're navigating a room, catching a ball, or simply walking down the street. The importance of vision in our everyday functioning is easy to overlook until it begins to fail. From entertainment to competition, modern life is saturated with visual information, reinforcing the idea that we are, fundamentally, visual creatures.

Nowhere is this reliance more apparent than in goal-directed activities like sports, where quick and accurate visual input helps us track opponents, locate teammates, and aim for a target. But what happens when vision is compromised–gradually or suddenly? Do we instinctively fall back on other systems, such as bodily awareness or memory? Do past experiences fill in the gaps left by fuzzy or absent visual cues?

Visual impairment is becoming increasingly common, with some projections estimating that half the global population will need corrective lenses by 2050 (Holden et al., 2016). This trend makes it essential to understand how people act and orient in space when vision can no longer be relied upon. My interest lies specifically in how proprioception–our internal sense of body position–and prior experience support spatial action under degraded visual conditions. If vision begins the process of mapping space, can proprioception maintain it?

This question is more than theoretical for me. I grew up playing ball sports like baseball and basketball with normal vision, but over the years my sight has worsened to the point where I now have 10/120 acuity. Recently, while playing inner tube water polo without glasses, I was still able to throw accurately–whether passing across the pool or aiming at the goal. What struck me was how natural it felt, even though I couldn't clearly see my teammates or the target. I found myself relying not solely on visual detail, but on years of ingrained movement patterns and my internal spatial representation of the pool. It made me wonder how others adapt in similar situations, and what this says about the ways we internalize and act within space.

This leads me to my central research question:

- How do people maintain internal spatial representations and perform goal-directed actions when visual input is impaired or unavailable?

To further explain this, I ask:

- To what extent can proprioception and motor memory compensate for the absence of visual cues in spatial tasks like throwing or navigation?
- How does prior experience with a space or task affect spatial confidence and accuracy under progressively degraded visual conditions?
- What does variation in performance and confidence across these conditions reveal about how internal spatial maps are constructed, maintained, and used?

Through this literature review, I aim to explore these questions by synthesizing work from cognitive science, perceptual psychology, and motor learning to elucidate conclusions. I will also propose an experimental design that builds on these insights to further investigate the relationship between experience, proprioception, and spatial performance without vision.

2 Vision and Internal Spatial Representations

2.1 Visual Contributions to Spatial Mapping

Vision plays a foundational role in how we build and update our internal understanding of space. It is not just a feed of raw input–it helps us extract structure, estimate distances, and make moment-to-moment decisions about where we are and how we should move. In spatial tasks like throwing or aiming, vision enables rapid calibration between the body and the environment, linking perception directly to action. But this reliance on vision also makes it a single point of failure. When it degrades, what happens to the internal models that guide movement?

From early development onward, visual information scaffolds the ability to form allocentric maps–representations that go beyond our current bodily position to encode the layout of the world in a stable, object-to-object framework (Newcombe et al., 2019). This mapping function is crucial not just for navigation, but for coordinating complex movement. Research comparing learning from maps versus direct navigation shows that visual overviews lead to more accurate internal spatial models, particularly when understanding the global structure of an environment is required (Thorndyke and Hayes-Roth, 1982). These findings imply that vision is especially useful when building high-level spatial schemas that need to be recalled later.

Still, vision is not the whole story. Evidence from blindfolded walking tasks shows that people can maintain and update spatial representations without any visual input at all (Rieser et al., 1992). These models, once formed, are not fragile—they can persist through movement, allowing people to track their position relative to unseen landmarks. Additionally, when vision is paired with translational movement, like walking through a space rather than just looking at it, people form stronger cognitive maps than when either modality is used alone (Ruddle et al., 2011, Rieser et al., 1992). These results complicate the idea that vision dominates spatial cognition. Instead, they suggest that the brain blends multiple systems—visual, proprioceptive, and vestibular—to build flexible internal models of space.

So while vision may initiate spatial representations, it is neither the sole author nor the most reliable long-term guide. Internal spatial maps persist beyond sight and can be strengthened through movement and experience. This opens the door to exploring what happens when visual access is compromised, and how other systems step in to preserve accurate action.

2.2 Navigating with Impaired Vision

Once visual input begins to degrade–whether through impairment, temporary occlusion, or environmental limits–spatial cognition becomes a test of how well those internal models hold up. In many cases, the shift is not immediate or catastrophic, but gradual and adaptive. The body learns to lean more on proprioception, vestibular feedback, and memory to stabilize performance.

When visual feedback is low-resolution or inconsistent, people struggle more to correct small spatial errors. For instance, adaptation to fine-grained sensorimotor discrepancies becomes more difficult, even though large-scale adjustments can still be made (Tsay et al., 2023). This suggests that while internal models may continue to function in broad strokes, precision suffers without reliable visual input. Even partial degradation, like the loss of one visual field, can measurably impair goal-directed movement (Sheppard et al., 2021), reinforcing the idea that vision remains a critical—but fragile—component of fine-tuned control.

That said, people are not helpless when vision falters. Engagement with the environment—moving through it, touching it, interacting physically–can reinforce spatial knowledge and reduce dependence on sight. Individuals who physically explore a space tend to form stronger internal representations than those who rely on vision alone (McCarthy et al., 2021). This aligns with the broader theme emerging in the literature: spatial competence is not purely perceptual, but deeply embodied.

This adaptability matters. As vision becomes less dependable, the brain appears capable of shifting weight toward other sensory systems. People may not be aware of this shift explicitly, but their actions reflect it. Accurate movement without sight isn't a trick of luck—it's a demonstration of how internal spatial models, when built through experience and refined by proprioception, can stand on their own. This capacity is at the heart of my research question: when vision recedes, how do we continue to act with spatial precision? To explore that, I now turn to the roles of proprioception and motor memory.

3 Proprioception, Motor Memory, and Action Without Vision

3.1 Proprioception and Sensorimotor Control

When vision is removed or degraded, proprioception becomes a crucial system for orienting in space and guiding goal-directed action. Often described as our "sixth sense," proprioception enables awareness of limb position, muscle tension, and movement without needing visual confirmation. This sensory system not only supports balance and coordination–it also plays a central role in building and updating internal spatial models during action.

Deficits in proprioception or vision expose just how interdependent these systems can be. Children with visual impairments, for example, perform worse on object control tasks like throwing or catching, likely due to underdeveloped proprioceptive systems that are not yet calibrated to compensate for missing visual input (Abdullah and Parnabas, 2014). Early disruption to broad-field vision can similarly impair the alignment of proprioceptive and visual signals, limiting the development of accurate nonvisual spatial representations (Rieser et al., 1992). At the same time, studies of rare individuals with near-total proprioceptive loss have shown that spatial performance can persist in surprising ways. One such individual (IW) maintained high accuracy in spatial tasks but showed longer reaction times, suggesting that the brain can compensate for missing sensory channels at the cost of efficiency (Renault et al., 2018). These examples highlight proprioception's value not just as backup but as an independent spatial contributor.

Proprioception also enables the calibration of motor responses to distortion. In a classic study, participants adapted their arm movements to visual perturbations through proprioceptive feedback alone–without direction-specific cues–suggesting that the brain modulates overall movement gain using internal body feedback (Bock, 1992). In this view, proprioception is not just reactive; it actively helps recalibrate the mapping between space and movement. These findings reinforce the idea that proprioception, even without vision, provides a stable basis for internal spatial guidance.

3.2 Motor Adaptation and Learned Experience

Beyond real-time feedback, prior experience can shape how accurately we act in space when vision is compromised. Motor learning–the process of refining action through repeated movement–helps solidify internal predictions about how movement maps onto external space. This learned calibration is especially critical when visual cues are no longer reliable.

Experience appears to buffer against visual loss, particularly when training directly targets the body's sensorimotor systems. Visually impaired athletes, for example, have improved upper-body control and throwing accuracy through structured proprioceptive training (Carretti et al., 2023, 2024). These gains reflect more than just physical skill–they suggest the brain can optimize alternative sensory systems when given the right practice environment.

This kind of sensorimotor adaptation is supported by internal models of movement outcomes, which are refined over time with consistent feedback (Bastian, 2008). Once these models are in place, they reduce the need for real-time visual confirmation. If someone has thrown a ball thousands of times in the same context, they no longer need to calculate each movement–they just act. Timely, predictive feedback further enhances this learning by aligning internal expectations with external results (Wang et al., 2024). This suggests that kinesthetic and proprioceptive information, when well-timed, can reinforce durable internal spatial representations.

Notably, people do not need large amounts of feedback to construct usable spatial models. Even minimal input–like a few visual-proprioceptive pairings–can lead to systematic generalizations. In one study, participants given sparse mappings still constructed accurate internal models by applying simple rules, often assuming linear relationships between input and output (Bedford, 1989). This ability to extract structure from limited exposure may explain how people can throw, aim, or move with precision even under degraded sensory conditions: their internal spatial model is based not just on observation, but on learned regularities from prior experience.

Taken together, these findings suggest that proprioception and motor memory are not merely fallback systems for when vision fails. They are core contributors to how we understand and act in space, and they can operate independently when trained through experience. This has direct implications for how we support individuals with visual impairments, but also for how we understand our own reliance on bodily systems in everyday spatial behavior.

4 Confidence, Experience, and Degraded Sensory Input

4.1 Metacognitive Judgments of Spatial Performance

Spatial cognition involves more than simply getting from one place to another accurately–it also involves confidence in the decisions we make along the way. Metacognition, or the ability to assess and reflect on our own performance, plays a key role in how people act under uncertainty. When vision becomes unreliable, individuals must depend on internal representations and past experience to guide their actions, making confidence a meaningful metric in how people evaluate and use those representations.

Studies show that people's confidence in spatial tasks can be influenced by both external conditions and individual strategies. For example, when participants used hand gestures during mental rotation tasks, their confidence improved along with their performance, especially under cognitively demanding conditions (Çapan et al., 2024). These findings suggest that embodied actions—like gesturing—can reinforce internal models and metacognitive trust. Age-related comparisons also reveal interesting patterns: even though older adults perform worse on some spatial tasks, their ability to monitor and evaluate their own spatial cognition tends to remain relatively intact (Ariel and Moffat, 2018). These results imply that spatial confidence is not entirely tied to raw performance; it can persist or even be trained separately.

However, degraded sensory input introduces new challenges for confidence. When visual information is unclear or unreliable, people may become less certain about their movements—even if those movements remain broadly effective. For instance, as mentioned earlier, individuals with low vision had difficulty adjusting to small-scale sensorimotor errors, likely due to limited feedback resolution (Tsay et al., 2023). Similarly, simulated low-vision conditions led to a noticeable decrease in motor precision and stability (Sheppard et al., 2021). Both studies point to a broader insight: when people lose access to clear feedback, their spatial confidence tends to falter, even when their internal models are functionally sufficient.

4.2 The Role of Experience and Familiarity

Despite these challenges, experience can act as a stabilizing force–both in terms of spatial accuracy and confidence. Familiarity with an environment or repeated engagement with a motor task builds a resilient internal model that can be relied upon when visual cues fade. This is especially apparent in individuals who have trained for years in vision-restricted contexts.

Structured training with proprioceptive feedback can improve both spatial performance and self-efficacy. Visually impaired athletes, for example, have shown improvements not just in accuracy and control but also in confidence after undergoing sensorimotor training programs (Carretti et al., 2023, 2024). These gains reflect how repeated practice can rewire spatial action systems and solidify belief in one's own capability. Similar effects have been observed across educational and training contexts, where embodied interaction with spatial problems leads to measurable gains in spatial cognition and task engagement (Lee-Cultura and Giannakos, 2020). A similar pattern appears in spatial performance research, where individuals born blind sometimes outperform those with early-onset partial vision—suggesting that consistent nonvisual input may support stronger compensatory strategies than degraded or unreliable visual experience (Rieser et al., 1992).

Experience does not exist in a vacuum—it is shaped by broader factors like gender roles and cultural expectations. Meta-analytic findings suggest that identification with masculine gender roles, regardless of biological sex, correlates positively with spatial ability (Reilly and Neumann, 2013). This may reflect differences in access to spatially intensive activities, encouragement to explore, or self-beliefs about one's own competence. From an evolutionary perspective, long-term behavioral differences in mobility and exploration may have led to sex-based variations in spatial cognition and confidence (Cashdan and Gaulin, 2016). These findings reinforce the idea that confidence in spatial tasks is deeply tied to the kinds of experience and socialization individuals accumulate over time.

Even at the sensory-motor level, repeated action builds not just accuracy, but assurance. With practice, internal models become more efficient, allowing individuals to predict and control movement outcomes even in the absence of visual input (Bastian, 2008; Wang et al., 2024). The more familiar a person is with the movement itself–through throwing, walking, or navigating–the more they can trust their internal sense of space to guide them when visual feedback is absent or degraded.

Altogether, the research points to a model of spatial cognition that is not just perceptual, but deeply experiential and metacognitive. Individuals who have trained in a space, who understand their own movement patterns, or who come from contexts that encourage spatial exploration are better equipped to maintain performance and confidence when vision fails. In this way, the relationship between sensory conditions, learned experience, and self-monitoring forms the foundation for resilient spatial action.

5 Discussion

This review set out to explore a simple but surprisingly underexamined question: How do people maintain internal spatial representations and perform goal-directed actions when vision is impaired or unavailable? More specifically, I asked to what extent proprioception and motor memory can compensate for reduced visual input, how prior experience influences spatial confidence and accuracy, and what this all suggests about the structure and flexibility of internal spatial models.

Across the literature, a consistent pattern emerges: vision is foundational, but not exclusive. It initiates and sharpens spatial representations, especially in early development and during tasks requiring large-scale layout comprehension. However, once spatial models are formed, they can be maintained, updated, and even used to guide precise action through non-visual systems. Proprioception, motor memory, and embodied experience provide stable inputs that can support navigation, throwing, and orientation under degraded sensory conditions. Confidence in these internal models varies depending on familiarity, feedback clarity, and the individual's history of spatial engagement–but it is not always tightly linked to accuracy. In many cases, people continue to act effectively even when they feel uncertain.

Still, important gaps remain. While many studies isolate one sense at a time, real-world spatial action almost always involves multisensory integration. Disentangling the specific contribution of proprioception, for example, is difficult without invasive methods or extremely rare cases like deafferentation. Likewise, simulating visual impairment with goggles or filters may not fully replicate the psychological and neurocognitive realities of long-term vision loss. These limitations make it hard to draw firm conclusions about how each modality operates in isolation versus together.

There are also theoretical tensions around how spatial models are constructed. Some research implies that spatial knowledge is hierarchical or allocentric, while others support more egocentric, action-based frameworks—especially under degraded conditions. Individual variation adds another layer of complexity. Factors like gender role identification, age, athletic experience, and cognitive style can all influence both performance and confidence, sometimes in unpredictable ways. These variables are often treated as confounds, but they may in fact be central to understanding how flexible and resilient spatial systems truly are.

What becomes clear is that internal spatial representation is not a fixed product of vision alone, but a dynamic process shaped by bodily engagement, feedback history, and metacognitive calibration.

Future work will need to take this complexity seriously, designing experiments that reflect the embodied, multisensory nature of spatial action. That includes investigating not just what people can do without vision, but how they feel about what they are doing—and how confidence itself becomes a functional part of the spatial reasoning system.

5.1 Future Work

To build on the findings of this review, I propose an experimental study that tests how individuals perform and self-evaluate goal-directed spatial actions under progressively degraded visual conditions. The focus is on understanding how proprioception, prior experience, and confidence interact when vision becomes less reliable or disappears altogether.

Participants will be asked to throw a soft object (such as a beanbag or paper ball) at a target after viewing and/or physically exploring the space. The key manipulation will involve four visual conditions presented within-subjects: (1) normal or corrected-to-normal vision, (2) simulated blurry vision, (3) blindfolded with prior visual exposure, and (4) blindfolded without any visual exposure–requiring reliance on proprioceptive and spatial memory alone. Across conditions, participants will also vary in their level of task familiarization: some will perform the task with no prior throws, while others will be allowed to train with visual feedback beforehand.

After each throw, participants will rate their confidence in their accuracy. Accuracy and confidence can then be compared across conditions to assess how performance and self-assessment change as visual information is removed and replaced by internal cues. This design allows for direct testing of how proprioceptive models support action under visual degradation, and whether confidence tracks with actual performance or diverges when sensory cues become ambiguous.

The study provides a flexible framework for examining real-time spatial cognition and adaptation. It can help clarify how internal representations are used and evaluated under uncertainty, and could inform training tools, assistive design, or rehabilitation strategies for individuals with visual impairments or sensory-motor mismatches.

6 Conclusion

Understanding how people act in space without vision reveals something deeper about how spatial cognition works as a whole. While sight is often the dominant modality in spatial tasks, it is far from the only one that matters. When vision is impaired or absent, the brain draws on a range of embodied resources—proprioception, motor memory, and accumulated experience–to stabilize performance. These systems do more than simply fill in gaps; they support an entire way of knowing the world that is grounded in movement, feedback, and familiarity.

Internal spatial representations can remain intact and functional without continuous visual input, especially when they are built through repeated experience. Confidence in those representations, though more fragile, can also persist if supported by metacognitive awareness and task familiarity.

From blindfolded navigation to visually impaired athletes executing complex throws, the evidence suggests that the body stores spatial knowledge in ways that do not always require seeing to act.

At the same time, many questions remain. How exactly are visual, proprioceptive, and vestibular inputs weighted and re-weighted as conditions change? What kinds of training most effectively strengthen proprioceptive spatial models? Can confidence in one's internal model be taught or adapted the same way accuracy can? These questions point to promising directions for future research, including the development of targeted training programs for individuals with visual impairments, the refinement of sensorimotor adaptation protocols, and the creation of assistive technologies that leverage bodily cues instead of relying solely on visual augmentation.

Ultimately, the ability to move and act with spatial precision–even without sight–demonstrates the adaptability of the human cognitive system. Experience and proprioception do not merely compensate for lost vision; they enable an alternate but equally valid path through space. Recognizing this opens new possibilities not just for rehabilitation and training, but for rethinking what it means to know where you are, where you're going, and how to get there.

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